Markov Chains (AI HL)

Section 1: Transition Matrices and State Vectors

1 In a standard transition probability matrix T (where entry T_{ij} represents the probability of moving from state i to state j), what must the sum of the entries in each row equal?

a. 1

b. 0

c. The number of states

d. $\frac{1}{n}$

2 Given initial state probability vector $\mathbf{s}_0 = [0.6, 0.4]$ and transition matrix $T = [[0.8, 0.2], [0.3, 0.7]]$, what is the state distribution after one step $\mathbf{s}_1 = \mathbf{s}_0 T$?

a. $[0.60, 0.40]$

b. $[0.55, 0.45]$

c. $[0.66, 0.34]$

d. $[0.48, 0.28]$

3 What is the Markov property (memoryless property) in stochastic processes?

a. The future state depends only upon the current state, not on the sequence of states that preceded it

b. The process has zero variance

c. All states have equal probability of occurrence

d. The chain can never visit the same state twice

Section 2: Multi-Step Transitions

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4 How is the probability distribution $\mathbf{s}_n$ after n transitions computed from the initial state vector $\mathbf{s}_0$ and transition matrix T?

a. $\mathbf{s}_n = \mathbf{s}_0 T^n$

b. $\mathbf{s}_n = \mathbf{s}_0 + nT$

c. $\mathbf{s}_n = T^n \mathbf{s}_0$

d. $\mathbf{s}_n = (\mathbf{s}_0 T)^n$

5 If $T = [[0.5, 0.5], [0.5, 0.5]]$, evaluate T^2.

a. $[[0.5, 0.5], [0.5, 0.5]]$

b. $[[0.25, 0.25], [0.25, 0.25]]$

c. $[[1, 0], [0, 1]]$

d. $[[0.5, 0], [0, 0.5]]$

6 What does entry $(T^3)_{2,1}$ represent?

a. The probability of transitioning from State 2 to State 1 in exactly 3 steps

b. The probability of transitioning from State 1 to State 2 in 3 steps

c. Three times the one-step probability

d. The cube of the probability from State 2 to State 1

Section 3: Steady-State (Stationary) Distribution

7 What defining matrix equation must the steady-state probability vector $\boldsymbol{\pi} = [\pi_1, \pi_2]$ satisfy?

a. $\boldsymbol{\pi} T = \boldsymbol{\pi}$ and $\sum \pi_i = 1$

b. $T\boldsymbol{\pi} = \mathbf{0}$

c. $\boldsymbol{\pi} T = \mathbf{0}$

d. $\boldsymbol{\pi} = T^{-1}$

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8 Find the stationary distribution for $T = [[0.7, 0.3], [0.2, 0.8]]$.

a. $\boldsymbol{\pi} = [0.4, 0.6]$

b. $\boldsymbol{\pi} = [0.5, 0.5]$

c. $\boldsymbol{\pi} = [0.3, 0.7]$

d. $\boldsymbol{\pi} = [0.6, 0.4]$

9 When is a Markov chain guaranteed to have a unique stationary distribution that is independent of the initial state?

a. When the transition matrix is regular (some power T^k has strictly positive entries)

b. Only if the matrix is symmetric

c. Whenever $\det(T) = 0$

d. Only if there are 2 states

Section 4: Absorbing Markov Chains

10 What is an absorbing state in a Markov chain?

a. A state that, once entered, cannot be left ($T_{ii} = 1$ and $T_{ij} = 0$ for $j \neq i$)

b. A state that has probability zero of being visited

c. A state that transitions with equal probability to all other states

d. A state where the steady-state probability is 0

11 In an absorbing Markov chain in canonical form $P = [[I, 0], [R, Q]]$, what does the fundamental matrix $N = (I - Q)^{-1}$ calculate?

a. The expected number of visits to each transient state before absorption

b. The probability of absorption in one step

c. The eigenvalues of the transition matrix

d. The total number of absorbing states

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12 How is the absorption probability matrix B calculated from fundamental matrix N and transient-to-absorbing matrix R?

a. $B = NR$

b. $B = RN$

c. $B = N + R$

d. $B = (I - Q)R$

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